But now the probability of getting something other than a six on one roll is $5/6$, and the probability that we don't get a six with m rolls is $(5/6)^m$ (we'll come back to this in the context of independence - in this example we're a bit sloppy with precision). That is

$$\mathbb{P}(C_{n,m}) = \left(\frac{5}{6}\right)^{m-n}$$

for all n, m. We have that

$$\mathbb{P}(B_n) = \lim_{m\to\infty} \left(\frac{5}{6}\right)^{m-n} = \left(\frac{5}{6}\right)^{-n} \underbrace{\lim_{m\to\infty} \left(\frac{5}{6}\right)^{m}}_{=0} = 0$$

and so

$$\mathbb{P}(A^c) = \lim_{n\to\infty} \mathbb{P}(B_n) = 0.$$

This was exactly our goal. Perhaps a self-evident statement, but still worth noting: event A is not the entire basic set Ω, but still its probability is 1. Probability theory says that it is almost certain to happen.

Week 1.

12\09\22

Ex 1:

(1) TRUE. $\{0\}$ is an elementary event

(2) FALSE. $0 \in A$ and $0 \in B$.

(3) FALSE.

$$\sum p_\omega = \sum_{i=1}^{\infty} \frac{1}{1+|\omega|} = \frac{1}{1+|0|} + \frac{1}{1+|1|} + \frac{1}{1+|-1|} + \dots$$

$$= 1 + 2\underbrace{\sum_{i=1}^{\infty} \frac{1}{1+i}}_{\text{this sum diverges.}}$$

Thus $\sum p_\omega \neq \mathbb{P}(\Omega) \neq 1$

(4) FALSE.

A valid probability distribution needs $\mathbb{P}(A) \geq 0$, and additionally (according to Th 3.1) $\mathbb{P}(A) \leq 1$.

(5) TRUE

According to Th 3.1.

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$$

$$= 0.1 + 0.2 - \mathbb{P}(A \cap B) \leq 0.3 < 0.4$$

(note that $\mathbb{P}(A \cap B) \geq 0$).

(6) TRUE.

$p_i = k$, where k is a constant ≥ 0.

If we let $k = 1/N$, we see that

$$\sum_{i=1}^{N} p_i = \sum_{i=1}^{N} (1/N) = N(1/N) = 1.$$

According to proposition 2.14

$$\mathbb{P}(A) = \sum_{i:\omega_i \in A} p_i$$ defines a probability distribution.

Since Ω is finite, any subset A is also finite.

$\Omega = \{\omega_1, \omega_2, \dots, \omega_N\}$.

$$\mathbb{P}(A) = \sum_{i:\omega_i \in A} p_i = \sum_{i:\omega_i \in A} 1/N = 1/N \sum_{i \in \omega_i \in A} 1$$

$$= 1/N \underbrace{\sum_{\omega_i \in \Omega} \mathbb{1}_A}$$

this counts how many elements there are inside A from the basic set Ω. Since $A \subseteq \Omega$, it actually counts the total number of elements of A

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Ex 2:

(1) $A \cap B \cap C$

(2) $A \cap B^c \cap C^c$

(3) We either have $A \cap B$ happening, $A \cap C$ happening, or $B \cap C$ happening:

$$A \cap B = (A \cap B \cap C) \cup (A \cap B \cap C^c)$$

$$A \cap C = (A \cap C \cap B) \cup (A \cap C \cap B^c)$$

$$B \cap C = (B \cap C \cap A) \cup (B \cap C \cap A^c)$$

$$(A \cap B) \cup (A \cap C) \cup (B \cap C)$$

$$= [(A \cap B \cap C) \cup (A \cap B \cap C^c)] \cup [(A \cap C \cap B) \cup (A \cap C \cap B^c)] \cup [(B \cap C \cap A) \cup (B \cap C \cap A^c)]$$

$$= (A \cap B \cap C) \cup (A \cap B \cap C^c) \cup (A \cap C \cap B^c) \cup (B \cap C \cap A^c)$$

Ex 3:

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Haj 3

(1) Before solving the exercise, I would like to change the notation p_ω to p_j, where p_j would represent the probability of event $\omega^{(j)}$, $j \in \mathbb{N}$. We can do this because the base set $\Omega = \{-1, 1\}^N$ is finite countable, meaning we can enumerate the set Ω with elements $\omega^{(j)} \in \Omega$, $j \in \mathbb{N}$. Thus

$$p_j = q^{\sum_{i=1}^{N} \frac{1+\omega_i^{(j)}}{2}} (1-q)^{\sum_{i=1}^{N} \frac{1-\omega_i^{(j)}}{2}}$$

For the exercise, we need to show that $p_j \geq 0$ for all $j \in \mathbb{N}$ and $\sum_{j:\, \omega^{(j)} \in \Omega} p_j = 1$.

First off, q is non-negative and $q \leq 1$ so $1-q \geq 0$. This implies that $p_j \geq 0$.

Next, we will show that $\sum_{j:\, \omega^{(j)} \in \Omega} p_j = 1$.

Notice that the exponents $\sum_{i=1}^{N} \frac{1+\omega_i^{(j)}}{2}$ and $\sum_{i=1}^{N} \frac{1-\omega_i^{(j)}}{2}$ are equal to the number of 1's and -1's in $\omega^{(j)}$, respectively.

Now, let n denote the number of 1's in $\omega^{(j)}$. Note that the number of -1's in $\omega^{(j)}$ is then $N-n$. We can rewrite p_j as

$$p_j = q^n (1-q)^{N-n}$$

where n is still the number of 1's in $\omega^{(j)}$.

It would be nice if we could sum over the n values, that is from $n=0$ to $n=N$. For $n=k$, that is $\omega^{(j)}$ contains k number of 1's, there are $\binom{N}{k}$ possibilities, that is N choose k elements.

$$\sum_{j=1}^{(2^N)} p_j = \sum_{n=0}^{n=N} \binom{N}{n} q^n (1-q)^{N-n} \overset{(\text{binom})}{=} 1$$

This is now a summation of a binomial distribution, which is known to be a probability distribution, implying that $\sum p_j = 1$.

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Haj 3

(2)

Let's analyse event A. If $\omega_1 + \omega_2 \cdots + \omega_N = N$, then each ω_j must be equal to 1. This means that the number of 1's is N which corresponds to one elementary event $\omega = (1, 1, 1, \cdots, 1)$.

$$\mathbb{P}(A) = \mathbb{P}(\{\omega\}) = q^N$$

Haj 4

We will verify the conditions of definition 2.9, which we denote by (1), (2) and (3) respectively.

(1) Does it satisfy $\forall A \subset \Omega$, $\mathbb{P}(A) \geq 0$?

Take $A \subset \Omega$

$\mathbb{P}(A) = \sum_{i=1}^{n} p_i \mathbb{P}_i(A)$, since $p_i \geq 0$ and $\mathbb{P}_i(A) \geq 0$ we have that $\mathbb{P}(A) \geq 0$.

(2) Does it satisfy $\mathbb{P}(\Omega) = 1$?

$$\mathbb{P}(\Omega) = \sum_{i=1}^{n} p_i \mathbb{P}_i(\Omega) = \sum_{i=1}^{n} p_i (1) = \sum_{i=1}^{n} p_i = 1.$$

(3) Does it satisfy, for $(A_i)_{i=1}^{\infty}$ disjoint sets, $\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i)$?

$$\mathbb{P}\left(\bigcup_{j=1}^{\infty} A_j\right) = \sum_{i=1}^{n} p_i \mathbb{P}_i\left(\bigcup_{j=1}^{\infty} A_j\right)$$

$$= \sum_{i=1}^{n} p_i \left(\sum_{j=1}^{\infty} \mathbb{P}_i(A_j)\right)$$

$$= \sum_{i=1}^{n} \sum_{j=1}^{\infty} p_i \mathbb{P}_i(A_j)$$

$$= \sum_{j=1}^{\infty} \sum_{i=1}^{n} p_i \mathbb{P}_i(A_j) = \sum_{j=1}^{\infty} \mathbb{P}(A_j)$$

We assumed here that we can change the order: $\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} = \sum_{j=1}^{\infty} \sum_{i=1}^{\infty}$.

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Hw 5

Let's analyse event A_0. If $-N$ is odd, then the event $w_1 + w_2 + \dots + w_N = 0$ cannot happen

$$w_1 + w_2 + \dots + w_{2k} + w_{2k+1} = 0$$

Let's prove by induction

For $k=1$

$w_1 = 0$ is not possible

Assume that it is not possible for $k=p$. Then

$$w_1 + \dots + w_{2p+1} \neq 0$$

For $k=p+1$,

$$w_1 + w_2 + \dots + w_{2p+1} + w_{2p+2} + w_{2p+3}$$

if $w_1 + w_2 + \dots + w_{2p+1} + w_{2p+2} + w_{2p+3} = 0$

then $w_1 + w_2 + \dots + w_{2p+1} = -(w_{2p+2} + w_{2p+3})$

and the RHS can take on values $-(1+1)$, $-(1-1)$, $-(-1+1)$ or $-(-1-1)$

Let's show that $w_1 + w_2 + \dots + w_{2k} = 2p$ where $p \in \mathbb{Z}$.

For $k=1$

$$w_1 + w_2 \in \{1+1, -1+1, +1-1, -1-1\}$$
$$= \{2, 0, 0, -2\} = \{-2, 0, 2\}.$$

Thus $w_1 + w_2 = 2p$, for some $p \in \mathbb{Z}$.

Assume true for $k = K$.

Then

$$w_1 + w_2 + \dots + w_{2K} = 2p.$$

For $k = K+1$ we have

$$(w_1 + w_2 + \dots + w_{2K}) + w_{2K+1} + w_{2K+2} = (2p) + w_{2K+1} + w_{2K+2}$$

and we have already shown that $w_{2K+1} + w_{2K+2} \in \{-2, 0, 2\}$

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and so $2p + \omega_{2k+1} + \omega_{2k+2} \in \{-2+2p, 2p, 2+2p\}$

where p is in $\mathbb{Z}$ from the inductive assumption

Thus, $\omega_1 + \dots + \omega_{2k} = 2p$, $p \in \mathbb{Z}$, for all $k \in \mathbb{N}$.

Thus, if $\omega_1 + \dots + \omega_{2k} + \omega_{2k+1} = 0$

then $-\omega_{2k+1} = \omega_1 + \dots + \omega_{2k} = 2p$

but $-\omega_{2k+1} \in \{-1, 1\}$ and there is no $p \in \mathbb{Z}$ such $1 = 2p$ or $-1 = 2p$. Thus, $\omega_1 + \dots + \omega_{2k+1} \neq 0$, meaning that for odd $N = 2k+1$, we have $\mathbb{P}(A) = 0$ because it is impossible.

Assume <u>N is even</u>. Then this means that there are the same number of 1's as there are -1's. This means that the exponents $\sum_{i=1}^{N} \frac{1-\omega_i}{2}$ and $\sum_{i=1}^{N} \frac{1+\omega_i}{2}$ are both equal to $N/2$ and so for an event $\omega \in A$.

$$p_\omega = q^{N/2}(1-q)^{N/2}.$$

Now, we notice that p_ω does not depend on ω, meaning that we simply need to calculate how many events like these are there in A. We have to choose $N/2$ elements out of a total of N elements ($N/2$ of the elements need to be a "1", the remaining values are forced to be "-1"). Using combinatorics this translates to $\binom{N}{N/2}$, N choose $N/2$. Let $M = \binom{N}{N/2}$

Thus $\mathbb{P}(A) = \mathbb{P}(\bigcup_{i=1}^{M} \omega_i)$. Each ω_i is an elementary event and hence disjoint meaning that

$$\mathbb{P}\left(\bigcup_{i=1}^{M} \omega_i\right) = \sum_{i=1}^{M} \mathbb{P}(\omega_i) = M(p_\omega) = \binom{N}{\frac{N}{2}} q^{N/2}(1-q)^{N/2}$$

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(6) If we have a finite non-empty basic set Ω with a uniform distribution, then the probability of event A depends only on the number of elements of A and the number of elements of Ω (and not, for example, which set A is in question).

Correct.

$p_i = k \quad \sum_{i:\omega_i \in \Omega} p_i = 1$ $\quad \omega_1, \{\omega_1, \omega_2\}$

$Np_i = 1, \; p_i = 1/N$

$\mathbb{P}(A) = 0 \iff A = \phi$

Ex 6: Probability calculus inequalities.

(1) Show that for any (not necessarily distinct) events $(A_j)_{j=1}^{n}$

$$\mathbb{P}\left(\bigcup_{j=1}^{n} A_j\right) \leq \sum_{j=1}^{n} \mathbb{P}(A_j).$$

Lemma 3.1 tells us that $\mathbb{P}(A \cup B) \leq \mathbb{P}(A) + \mathbb{P}(B)$.

Assume true for $n = k$

$$\mathbb{P}\left(\bigcup_{j=1}^{k} A_j\right) \leq \sum_{j=1}^{k} \mathbb{P}(A_j)$$

then for $n = k+1$

$$\mathbb{P}\left(\bigcup_{j=1}^{k+1} A_j\right) = \mathbb{P}\left(\left(\bigcup_{j=1}^{k} A_j\right) \cup A_{k+1}\right)$$

using base case

$$\leq \mathbb{P}\left(\bigcup_{j=1}^{k} A_j\right) + \mathbb{P}(A_{k+1})$$

using assumption

$$\leq \sum_{j=1}^{k} \mathbb{P}(A_j) + \mathbb{P}(A_{k+1})$$

$$= \sum_{j=1}^{k+1} \mathbb{P}(A_j) \quad \square$$

Thus, by induction the statement is true for all natural numbers greater than 2 and also including 1.

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Ex 6

(2) Show that for any (not necessarily disjoint) events $(A_j)_{j=1}^{\infty}$

$$\mathbb{P}\left(\bigcup_{j=1}^{\infty} A_j\right) \leq \sum_{j=1}^{\infty} \mathbb{P}(A_j)$$

[Hint: you can try to approach the problem in two ways: you can either try to justify (using monotonicity) why the limit $n \to \infty$ can be taken from the result in the first part, or you can try to write the event $\bigcup_{j=1}^{\infty} A_j$ as a combination of separate events (good events for this could be $B_1 = A_1$ and $B_n = A_n \setminus \bigcup_{j=1}^{n-1} A_j$ when $n \geq 2$), and estimate the corresponding probabilities.]

$B_1 = A_1$, $B_2 = A_2 \setminus A_1$, $B_3 = A_3 \setminus \bigcup_{j=1}^{2} A_j$

$$\bigcup_{j=1}^{\infty} B_j = \bigcup_{j=1}^{\infty} A_j \setminus \left(\bigcup_{k=1}^{j-1} A_k\right) = \bigcup_{j=1}^{\infty} A_j$$

$\bigcup_{j=1}^{\infty} B_j$ is a union of disjoint events so by definition 2.9. of probability distributions

$$\mathbb{P}\left(\bigcup_{j=1}^{\infty} B_j\right) = \sum_{j=1}^{\infty} \mathbb{P}(B_j)$$

$$= \sum_{j=1}^{\infty} \mathbb{P}\left(A_j \setminus \bigcup_{k=1}^{j-1} A_k\right)$$

According to (lemma 3.1 (2)) we have that; letting $B_j = A_j \setminus \bigcup_{k=1}^{j-1} A_k$:

$$\mathbb{P}(B_j) = \mathbb{P}\left(A_j \setminus \bigcup_{k=1}^{j-1} A_k\right) = \mathbb{P}(A_j) - \mathbb{P}\left(A_j \cap \bigcup_{k=1}^{j-1} A_k\right)$$

$$\mathbb{P}\left(\bigcup_{j=1}^{\infty} B_j\right) = \sum_{j=1}^{\infty} \mathbb{P}\left(A_j \setminus \bigcup_{k=1}^{j-1} A_k\right) = \sum_{j=1}^{\infty} \left(\mathbb{P}(A_j) - \mathbb{P}\left(A_j \cap \bigcup_{k=1}^{j-1} A_k\right)\right)$$

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$$= \lim_{n\to\infty} \sum_{j=1}^{n} \left(\mathbb{P}(A_j) - \mathbb{P}(A_j \cap \bigcup_{k=1}^{j-1} A_k) \right)$$

$$= \lim_{n\to\infty} \left(\left(\sum_{j=1}^{n} \mathbb{P}(A_j) \right) - \sum_{j=1}^{n} \mathbb{P}(A_j \cap \bigcup_{k=1}^{j-1} A_k) \right)$$

We assume here that $\lim_{n\to\infty} \sum_{j=1}^{n} \mathbb{P}(A_j)$ exists as well as $\lim_{n\to\infty} \sum_{j=1}^{n} \mathbb{P}(A_j \cap \bigcup_{k=1}^{j-1} A_k)$. Thus, we get the following:

$$= \sum_{j=1}^{\infty} \mathbb{P}(A_j) - \sum_{j=1}^{\infty} \mathbb{P}(A_j \cap \bigcup_{k=1}^{j-1} A_k) \leq \sum_{j=1}^{\infty} \mathbb{P}(A_j).$$

where $\sum_{j=1}^{\infty} \mathbb{P}(A_j \cap \bigcup_{k=1}^{j-1} A_k) \geq 0$ since $\mathbb{P}(A_j \cap \bigcup_{k=1}^{j-1} A_k) \geq 0$.

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